

The trace identity $D(q) = \text{cross}_{(3,1,1) \rightarrow (3,2)}(q)$:
structural reduction, computational verification, and the residual
gap

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Abstract

`2026-05-06-paths-explicit-and-Q3-refuted.tex` reported a computationally verified, structurally unexplained identity: the “ s_5 -step diagonal” contribution $D(q)$ to $\sigma_1(V_{(3,2,1)})$ at S_6 equals the cross-block trace $\text{cross}_{(3,1,1) \rightarrow (3,2)}(q)$ in the KL basis. Both equal $q^5(1+q)(q^4 + 7q^3 + 12q^2 + 7q + 1)$. After tightening the convention (the path-enumeration code uses the C_w basis, where D_i records left *ascents*, not descents), we reduce the identity to a clean residual:

$$\text{cross}_{(3,1,1) \rightarrow (3,2)}(q) = (1+q)(T_A(q) + T_B(q)),$$

where $T_A := \text{tr}(\Pi^{S_5} P_{(2,2,1)} R'_5)$ and $T_B := \text{tr}(\Pi^{S_5} D_5^{(3,1,1)\text{asc}} R'_5)$. We verify the residual computationally and obtain the surprising explicit values $T_A = q^6(1+q)^2$ and $T_B = q^5((1+q)^4 + 2q(1+q)^2)$. The factorizations are clean enough to suggest a structural explanation, but we do not provide one. The remaining gap: a path-level (or representation-theoretic) reason for these specific factorizations.

1 Setup and convention

We work inside the framework of `2026-04-24-sigma1-G1.tex` and the May 6 papers, but with one important sharpening of convention.

KL basis and the operator $T_i + 1$

Let $H_q = H_q(\mathfrak{S}_6)$ act on the cell module $V := V_{(3,2,1)}$ on the unprimed Kazhdan–Lusztig basis $\{C_w : w \in \mathcal{C}_{(3,2,1)}\}$ indexed by SYTs of shape $(3, 2, 1)$. In this basis,

$$T_i C_w = \begin{cases} -C_w & \text{if } s_i \in D_L(w) \text{ (descent),} \\ q C_w + \sum_{w' < w, s_i \in D_L(w')} \mu(w', w) C_{w'} & \text{if } s_i \notin D_L(w) \text{ (ascent),} \end{cases}$$

so that

$$T_i + 1 = (1+q) D_i + v M_i, \tag{1}$$

with

- D_i the diagonal $\{0, 1\}$ -matrix recording left *ascents*: $(D_i)_{ww} = 1$ iff $s_i \notin D_L(w)$;
- M_i the off-diagonal matrix with entries $(M_i)_{w', w} = \mu(w', w)$ for KL edges $\{w, w'\}$ where s_i is descent at one and ascent at the other; $v = \sqrt{q}$.

Note. This is the convention used in the path-enumeration code (`wgraph_fast.py`, `matrix_D.i`, `operator_T_plus_1_numeric`), which differs from the convention used in the prose of 2026-05-06-theorem-B-path but is internally consistent and recovers the correct $\sigma_1(V_\lambda)$ values.

The staircase factorization

Fix the S_6 staircase reduced word for w_0 :

$$\underline{w}_0 = (1; 2, 1; 3, 2, 1; 4, 3, 2, 1; 5, 4, 3, 2, 1), \quad N = \binom{6}{2} = 15.$$

Positions 1–10 form the S_5 staircase; positions 11–15 are the new block $(s_5, s_4, s_3, s_2, s_1)$. Define

$$\begin{aligned} \Pi^{S_5} &:= \prod_{j=1}^{10} (T_{i_j} + 1), \\ R_5 &:= (T_5 + 1)(T_4 + 1)(T_3 + 1)(T_2 + 1)(T_1 + 1), \\ R'_5 &:= (T_4 + 1)(T_3 + 1)(T_2 + 1)(T_1 + 1). \end{aligned}$$

So $\Pi_q := \Pi^{S_5} \cdot R_5 = \Pi^{S_5} \cdot (T_5 + 1) \cdot R'_5$.

The two splits of $\sigma_1(V)$

Both decompositions are recalled from the May 6 papers (`wgraph_fast.py`, `s5_block.py`):

- **s_5 -step (Q3-refuted) split.** Decomposing $T_5 + 1$ at position 11 via (??),

$$\sigma_1(V) = D(q) + G(q),$$

where

$$\begin{aligned} D(q) &:= (1 + q) \operatorname{tr}(\Pi^{S_5} D_5 R'_5) = q^5 + 8q^6 + 19q^7 + 19q^8 + 8q^9 + q^{10}, \\ G(q) &:= v \operatorname{tr}(\Pi^{S_5} M_5 R'_5) = q^4 + 15q^5 + 73q^6 + 153q^7 + 153q^8 + 73q^9 + 15q^{10} + q^{11}. \end{aligned}$$

- **KL-block cross-trace split.** The branching $V \downarrow S_5 = V_{(3,2)} \oplus V_{(3,1,1)} \oplus V_{(2,2,1)}$ partitions the 16 SYTs of $(3, 2, 1)$ by the row of the box containing 6. Let P_μ be the diagonal projector onto the KL index-block V_μ . Define

$$\text{cross}_{\mu_i \rightarrow \mu_j}(q) := \operatorname{tr}(P_{\mu_i} \Pi^{S_5} P_{\mu_j} R_5).$$

Then $\sigma_1(V) = \sum_{\mu_i, \mu_j} \text{cross}_{\mu_i \rightarrow \mu_j}$. *Critical fact* (`s5_block_decomp.txt`): of the 9 cross-traces, only 5 are nonzero. The 4 vanishing crosses are $(3, 2) \rightarrow (3, 1, 1)$, $(3, 2) \rightarrow (2, 2, 1)$, $(3, 1, 1) \rightarrow (2, 2, 1)$, $(2, 2, 1) \rightarrow (3, 1, 1)$.

The descent structure for s_5

For SYT T of shape $(3, 2, 1)$, $s_5 \in D_L(T)$ iff $\text{row}(6) > \text{row}(5)$. Hence:

- $V_{(3,2)}$ -block (6 in row 2, 5 SYTs at indices $\{0, 2, 4, 8, 10\}$): *always* a descent.
- $V_{(3,1,1)}$ -block (6 in row 1, 6 SYTs at indices $\{1, 3, 6, 9, 12, 14\}$): descent iff 5 in row 2 (3 SYTs at $\{6, 12, 14\}$); ascent iff 5 in row 0 (3 SYTs at $V_{(3,1,1)}^{\text{asc}} = \{1, 3, 9\}$).

- $V_{(2,2,1)}$ -block (6 in row 0, 5 SYTs at indices $\{5, 7, 11, 13, 15\}$): *never* a descent (always ascent). Since D_5 in our convention is the *ascent* indicator,

$$D_5 = P_{(2,2,1)} + D_5^{(3,1,1)\text{asc}}, \quad (2)$$

where $D_5^{(3,1,1)\text{asc}}$ is the projector onto $V_{(3,1,1)}^{\text{asc}} = \{1, 3, 9\}$. In particular $P_{(3,2)}D_5 = 0$ (the $V_{(3,2)}$ -block lies entirely in the descent locus where D_5 vanishes).

2 Statement of the trace identity

Theorem 1 (Trace identity, computationally verified).

$$D(q) = \text{cross}_{(3,1,1) \rightarrow (3,2)}(q) = q^5 + 8q^6 + 19q^7 + 19q^8 + 8q^9 + q^{10} = q^5(1+q)(q^4 + 7q^3 + 12q^2 + 7q + 1).$$

The verification is in `wgraph_fast.py` (Lagrange interpolation, 18 evaluations on $V_{(3,2,1)}$) and `s5_block.py` (cross-trace decomposition). What we attempt below: a *structural* reduction.

3 Structural reduction

We use the convention-corrected decompositions to expand both sides.

LHS expansion

By (1),

$$\begin{aligned} D(q) &= (1+q) \text{tr}(\Pi^{S_5} (P_{(2,2,1)} + D_5^{(3,1,1)\text{asc}}) R'_5) \\ &= (1+q) [T_A + T_B], \end{aligned} \quad (3)$$

where

$$T_A := \text{tr}(\Pi^{S_5} P_{(2,2,1)} R'_5), \quad T_B := \text{tr}(\Pi^{S_5} D_5^{(3,1,1)\text{asc}} R'_5). \quad (4)$$

RHS expansion

Using $R_5 = (T_5 + 1)R'_5 = ((1+q)D_5 + vM_5)R'_5$ and the fact that $P_{(3,2)}D_5 = 0$:

$$\begin{aligned} \text{cross}_{(3,1,1) \rightarrow (3,2)}(q) &= \text{tr}(P_{(3,1,1)} \Pi^{S_5} P_{(3,2)} R_5) \\ &= (1+q) \text{tr}(P_{(3,1,1)} \Pi^{S_5} P_{(3,2)} D_5 R'_5) + v \text{tr}(P_{(3,1,1)} \Pi^{S_5} P_{(3,2)} M_5 R'_5) \\ &= 0 + v \text{tr}(P_{(3,1,1)} \Pi^{S_5} P_{(3,2)} M_5 R'_5) \\ &= v \text{tr}(P_{(3,1,1)} \Pi^{S_5} P_{(3,2)} M_5 R'_5). \end{aligned} \quad (5)$$

The first equality used cyclicity of trace; the third used $P_{(3,2)}D_5 = 0$. *This is the convention-correction's payoff*: in the C_w basis with D_5 as the ascent indicator, the entire diagonal piece of $\text{cross}_{(3,1,1) \rightarrow (3,2)}$ vanishes, leaving only the off-diagonal vM_5 contribution.

The residual identity

Setting $D = \text{cross}_{(3,1,1) \rightarrow (3,2)}$ and combining (??) and (??):

Theorem 2 (Residual identity). *Theorem 1 is equivalent to*

$$\boxed{v \operatorname{tr}(P_{(3,1,1)} \Pi^{S_5} P_{(3,2)} M_5 R'_5) = (1+q)(T_A + T_B)} \quad (6)$$

This is striking. The LHS sums over closed paths starting in $V_{(3,1,1)}$, traversing Π^{S_5} to a vertex in $V_{(3,2)}$, taking the (forced) M -step at position 11 to a vertex outside $V_{(3,2)}$ (since M_5 moves between s_5 -asymmetric pairs and $V_{(3,2)}$ is descent), then closing via R'_5 .

The RHS counts closed paths where step 11 acts as the $(1+q)$ -multiplier on $V_{(2,2,1)}$ (term T_A) or on $V_{(3,1,1)}^{\text{asc}}$ (term T_B).

Path-level disjointness. The two sides count over disjoint path multisets: LHS paths have $w_{10} = w_{15} = w_0 \in V_{(3,1,1)}$ and step 11 = M -step exiting from $V_{(3,2)}$; RHS paths have step 11 = D -step at a vertex in $V_{(2,2,1)} \cup V_{(3,1,1)}^{\text{asc}}$. The polynomial identity holds despite the path-set disjointness.

4 Computational verification of the residual

We compute all four traces directly. Script: `~/projects/scratch/2026-05-07-trace-identity/residual_check`
Lagrange interpolation over $q = 0, 1, \dots, 14$ (15 points).

Theorem 3 (Explicit formulas).

$$T_A = q^6 + 2q^7 + q^8 = q^6(1+q)^2, \quad (7)$$

$$T_B = q^5 + 6q^6 + 10q^7 + 6q^8 + q^9 = q^5((1+q)^4 + 2q(1+q)^2), \quad (8)$$

$$T_A + T_B = q^5(1 + 7q + 12q^2 + 7q^3 + q^4), \quad (9)$$

$$v \operatorname{tr}(P_{(3,1,1)} \Pi^{S_5} P_{(3,2)} M_5 R'_5) = q^5 + 8q^6 + 19q^7 + 19q^8 + 8q^9 + q^{10}, \quad (10)$$

$$(1+q)(T_A + T_B) = q^5 + 8q^6 + 19q^7 + 19q^8 + 8q^9 + q^{10}. \quad (11)$$

(??) and (??) agree as polynomials in q , confirming the residual identity Theorem 2, and hence Theorem 1.

Computational proof. The four traces are computed numerically by evaluating the matrix products at $q = 0, \dots, 14$, then Lagrange-interpolating. Polynomial multiplication (and the multiplication by $(1+q)$) is exact integer arithmetic. The output of `residual_check.py`:

```
T_A      = q^6 + 2q^7 + q^8
T_B      = q^5 + 6q^6 + 10q^7 + 6q^8 + q^9
LHS      = q^5 + 8q^6 + 19q^7 + 19q^8 + 8q^9 + q^10
(1+q)(T_A + T_B) = q^5 + 8q^6 + 19q^7 + 19q^8 + 8q^9 + q^10
Equal? True
```

The verification confirms the identity at the level of polynomial coefficients; this is decisive (a polynomial of degree 10 is determined by 11 values, and we verified 15). $\square$

The structure of the explicit values is suggestive

Two observations on $(??)$ – $(??)$ stand out.

Remark 4 ($T_A = q^6(1+q)^2$ is a single σ_1 -G1 atom). T_A has the cleanest possible form: a monomial in q times $(1+q)^{2k}$, sitting in the canonical basis $B_{2,4}$. In the language of 2026-04-24-sigma1-G1.tex, this means *every closed path starting in $V_{(2,2,1)}$ at step 11 has the same bigrade modulo the σ_1 -G1 prefix*. The bigrade is concentrated at a single $\bar{c}$ -value.

This is a strong structural statement. It says: among all closed 14-step paths in $V_{(3,2,1)}$ along the staircase word *minus position 11*, with the constraint $w_{10} = w_{11} \in V_{(2,2,1)}$, every such path has bigrade $(c, d) = (6, 2)$ in the unscaled count. The number of such paths is $T_A(1) = 1 + 2 + 1 = 4$.

Remark 5 ($T_B = q^5 \cdot \text{atom with } B_4\text{-decomp } (1, 2, 0)$). The factor $(1+q)^4 + 2q(1+q)^2$ is a B_4 atom with multiplicity vector $(1, 2, 0)$. Note the absence of the q^2 term. In path-counting language, paths constrained to $w_{10} = w_{11} \in V_{(3,1,1)}^{\text{asc}}$ never reach the highest bigrade $\bar{c} = 2$. Together with T_A :

$$T_A + T_B = q^5(1 + 7q + 12q^2 + 7q^3 + q^4).$$

The inner palindromic factor $1 + 7q + 12q^2 + 7q^3 + q^4$ has B_4 -decomp $(1, 3, 0)$. So:

$$T_A + T_B \text{ has } B_4\text{-decomp } (1, 3, 0).$$

Remark 6 (The decomposition $T_A + T_B$ vs. $A_{(4,1)}$). Compare $A_{(4,1)} = 1 + 7q + 17q^2 + 7q^3 + q^4$, B_4 -decomp $(1, 3, 5)$. The polynomial $T_A + T_B = q^5 \cdot A^*$ where $A^* = 1 + 7q + 12q^2 + 7q^3 + q^4$, B_4 -decomp $(1, 3, 0)$. So

$$A^* = A_{(4,1)} - 5q^2.$$

The B_4 -decomp differs by $(0, 0, 5)$, i.e., $5q^2$. This is exactly $5 = m_{(4,1)}^{(2)}$, the multiplicity of the highest- $\bar{c}$ atom in $V_{(4,1)}$. So $T_A + T_B$ is “ $A_{(4,1)}$ minus its top atom”. *This is suggestive but not yet a proof.*

5 Vanishing of the four cross-traces is structural

We re-state and partially explain the vanishing of the 4 cross-traces, since these underlie the structural reduction.

Lemma 7 (Cross-flow always into $V_{(3,2)}$). $\text{cross}_{\mu_i \rightarrow \mu_j}(q) = 0$ *identically as a polynomial in q whenever $\mu_j \neq (3, 2)$ and $\mu_i \neq \mu_j$. The 4 vanishing crosses are exactly $(3, 2) \rightarrow (3, 1, 1)$, $(3, 2) \rightarrow (2, 2, 1)$, $(3, 1, 1) \rightarrow (2, 2, 1)$, $(2, 2, 1) \rightarrow (3, 1, 1)$.*

Proof sketch. Each $\text{cross}_{\mu_i \rightarrow \mu_j}$ is a sum of products $(\Pi^{S_5})_{ij}(R_5)_{ji}$ with $i \in V_{\mu_i}, j \in V_{\mu_j}$. Each summand is a product of two non-negative-coefficient polynomials in q (in $a + bq$ form with $a, b \in \mathbb{Z}_{\geq 0}[q]$), hence non-negative. The sum is zero iff each summand is zero. So Lemma 3 is the conjunction of $|V_{\mu_i}| \cdot |V_{\mu_j}|$ scalar identities of the form “ $(\Pi^{S_5})_{ij} = 0$ or $(R_5)_{ji} = 0$ ”.

(Stated, not proved structurally here.) The vanishing has been verified by explicit computation in `s5_block.py`. A structural proof presumably uses (i) length-parity bookkeeping in the W -graph (KL edges connect vertices of opposite length parity), and (ii) the staircase-word reduced expression’s specific descent/ascent structure. $\square$

6 The gap

What remains for a clean structural proof?

1. **Structural reason for $T_A = q^6(1+q)^2$.** Why does the constraint $w_{10} \in V_{(2,2,1)}$ force every closed staircase-minus-step-11 path to have the *same* bigrade $(c, d) = (6, 2)$? A path-counting argument should give this: enumerate the closed paths with $w_{10} \in V_{(2,2,1)}$, observe each has the constrained bigrade, count $1 + 2 + 1 = 4$ paths.

The $V_{(2,2,1)}$ -block has 5 vertices, all with D_L containing $\{4, 5\}$ as ascents (since 5 below 6 means $s_5 \notin D_L$; 4 below 5 in row 0 plus 5 in row 1+ means $s_4 \notin D_L$ etc.). The W -graph restriction to $V_{(2,2,1)}$ is small, and *constraints from the staircase word* (which steps $\notin S$ at fixed staircase positions) likely force the bigrade.

2. **Structural reason for $T_B = q^5((1+q)^4 + 2q(1+q)^2)$.** The B_4 -decomp $(1, 2, 0)$ is the same shape as a sub-structure of $V_{(3,1,1)}$. Specifically, $V_{(3,1,1)}^{\text{asc}} = \{1, 3, 9\}$ is a subset of 3 vertices of $V_{(3,1,1)}$ (which has 6 vertices total). The W -graph restricted to these 3 vertices has *some* structure, and T_B counts paths constrained to land here at step 11.

A natural conjecture: $T_B = q \cdot \sigma_1^{V_{(3,1,1)}^{\text{asc}}}$ (some contracted W -graph). We do not pursue this.

3. **Path-bijective proof.** The polynomial identity $D = \text{cross}$ holds despite path-set disjointness (Section 3). A bijection between the disjoint path sets $\{D\text{-paths}\}$ and $\{\text{cross-paths}\}$ would have to be *cross-cutting*: it cannot send paths to paths preserving step 11's character (D vs M). Most natural would be a path-rewriting move that swaps a D-step at position 11 with an M-step plus some compensating local change. Such moves exist in W -graph theory (the “Knuth equivalences” on reduced words), but adapting them here is not immediate.
4. **The relation to $A_{(4,1)}$.** Remark ?? flags that $T_A + T_B = q^5 \cdot (A_{(4,1)} - 5q^2)$. The 5 here is suggestive: $5 = m_{(4,1)}^{(2)}$ = the multiplicity of the highest-grade atom in the σ_1 -G1 decomposition of $A_{(4,1)}$. *Whether $T_A + T_B$ truly is “ $A_{(4,1)}$ at S_5 minus its top atom” is open.* If true, it would tie the trace identity directly to $V_{(4,1)}$, completing the structural picture.

7 Honest assessment

What this paper achieves.

1. A convention-corrected reduction of Theorem 1 to a single residual identity (4) of the form $\text{cross} = (1+q)(T_A + T_B)$.
2. Explicit computation of T_A, T_B as polynomials in q , with clean closed forms (Theorem ??).
3. Identification of two structurally suggestive features: T_A is a single σ_1 -G1 atom, and $T_A + T_B$ is “ $A_{(4,1)}$ minus its top atom” (Remarks ??, ??).
4. A computational verification (`residual_check.py`) confirming both Theorem 1 and Theorem 2.

What this paper does *not* achieve.

1. A structural proof of Lemma 3 (the vanishing of 4 cross-traces).
2. A structural proof of $T_A = q^6(1+q)^2$: this is checked numerically but not derived from W -graph structure on $V_{(2,2,1)}$.

3. A structural proof of $T_B = q^5((1+q)^4 + 2q(1+q)^2)$: same.
4. A structural reason for the relation $A^* = A_{(4,1)} - 5q^2$ (Remark ??).
5. A path-level bijection between D -paths and cross-paths (or a sign-reversing involution on the symmetric difference).

Status. Theorem 1 is reduced to two finite path-counting facts (T_A and T_B explicit formulas), each of which is small enough to be verifiable by direct enumeration. The structural payoff is that Theorem ?? replaces a 6-coefficient surprise polynomial identity with two 4-coefficient sub-identities, each of clean form. The path is open: enumerate the bigrades of paths in $V_{(2,2,1)}$ and $V_{(3,1,1)}^{\text{asc}}$ along the staircase-minus-position-11.

Acknowledgments and references

Computational data: `~/projects/scratch/2026-05-06-paths/` (May 6, 2026), `~/projects/scratch/2026-05-06-paths/` (May 7, 2026, this session). Companion papers: `2026-05-06-twin-multiset.tex`, `2026-05-06-theorem-B-multiset.tex`, `2026-05-06-theorem-B-paths.tex`, `2026-05-06-paths-explicit-and-Q3-refuted.tex`.

Convention note. The path-enumeration code uses the unprimed C_w KL basis where D_i records ascents. The May 6 prose papers `theorem-B-paths` and `paths-explicit-and-Q3-refuted` say “ D_i records descents” which is the opposite convention. Both conventions give the same polynomial σ_1 values (sum over closed paths is symmetric under the basis swap), but local algebraic identities like $P_{(3,2)}D_5 = 0$ vs. $P_{(3,2)}D_5 = P_{(3,2)}$ differ. The present paper uses the code’s convention throughout.